

CÓDIGO:

Question 1. Chemical periodicity (5% of the total; only 10th and 11th grades)

1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	Total	Total
2	2	2	2	2	2	2	2	2	2	20	5%

Complete the following statements.

Please do not consider the chemical elements of period 7 in your analysis since these are radioactive or artificial and, due to their high instability, their chemistry has been little studied.

1.1. The chemistry of the element Si has certain analogies with the chemistry of the element C. However, Si compounds have larger bond distances because the _____ of Si is larger. Furthermore, the chemical bonds of Si with F, O and N are more polar because the _____ of Si is lower.

1.2. Period ____ elements have unusually low electron affinities due to their small atomic radius and high electron density.

1.3. The most basic hydride of the stable alkali metals is _____.

1.4. The superior oxide of the elements of group 14 with the greatest oxidizing character is _____.

1.5. The highest secondary ionization energy in the first transition series is possessed by the element _____.

1.6. The maximum possible oxidation state in the first transition series is not reached beyond the group _____.

1.7. The volatile hydrogenated compound with the highest boiling temperature among those formed by the elements of group 16 is _____.

1.8. The cations Cu^+ , K^+ and Cu^{2+} are arranged in increasing order of ionic radius according to _____.

1.9. The only acidic hydroxide (non-amphoteric) formed by group 13 (IIIA) elements is _____.

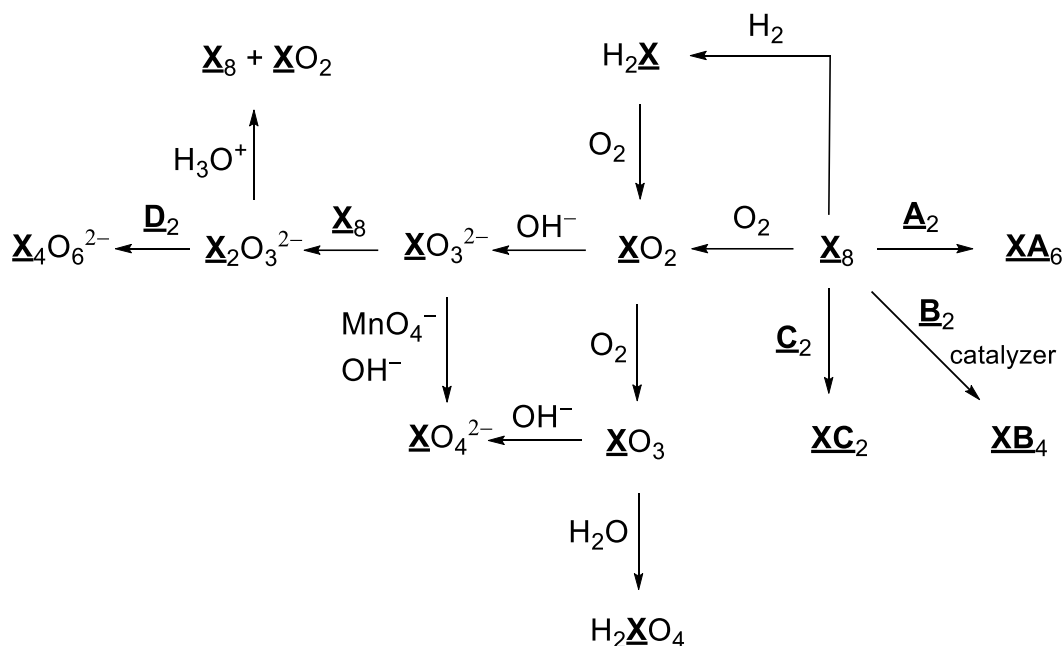
1.10. The binding energy of fluorine is relatively _____ among halogens even though the element fluorine has the smallest atomic radius of all the group 17 elements.

Question 2. “Here it smells to ...” (20% of the total; only 10th and 11th grades)

2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	2.10	Total	Total
10	10	4	24	4	6	2	4	6	10	80	20%

The following diagram shows some of the chemical properties of the compounds of the chemical element **X**, whose simple substance **X₈** is a yellow solid with a characteristic odor that has medicinal properties. The elements **A**, **B**, **C** and **D** are in the same group of the periodic table and the oxidizing power of their simple substances **A₂**, **B₂**, **C₂** and **D₂** decreases in that exact order.

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- 2.1. Say the identity of the chemical elements represented by letters.
- 2.2. Propose a mechanism for the reaction between the oxide XO_2 and C_2 in aqueous solution, forming the acids H_2XO_4 and HC . Formulate the formation of “logical” intermediaries.
- 2.3. Create chemical test equations to identify species X^{2-} y XO_4^{2-} in the laboratory.
- 2.4. Represent the structures of the species H_2X , XA_6 , XB_4 , XC_2 , XO_2 , XO_3 , H_2XO_4 and $\text{X}_4\text{O}_6^{2-}$. For the first four, indicate the molecular geometry.
- 2.5. Nitric acid in the presence of H_2XO_4 forms the nitronium cation NO_2^+ . Represent the resonant structures of this ion and classify it according to the Theory of Hard and Soft Acids and Bases (TABDB).
- 2.6. State the adjusted ionic equation for the oxidation of XO_3^{2-} with permanganate in a basic medium.
- 2.7. Compare the acid-base character of the oxides XO_2 and XO_3 based on the partial charges on the atoms in the molecules.
- 2.8. The anions $\text{X}_2\text{O}_3^{2-}$ and XO_4^{2-} are isoelectronic, however, the former reacts with D_2 and the latter does not react. Justify this fact according to the postulates of the TABDB.
- 2.9. Represent the electronic distribution of the XO_2 molecule using the Valencia Bond Theory. **Take into account the structure you drew in section 2.4.** Use only hybrid orbitals to describe the electrons of the central atom.
- 2.10. XO_3^{2-} formation from XO_2 in a basic medium occurs in two steps. First, the $\text{XO}_2\cdot\text{OH}^-$ adduct is formed and then deprotonated by a hydroxide ion. Consider that the external electronic structure by molecular orbitals of XO_2 is $(\dots) (\pi^b)^2 (\pi^n)^4 (\pi^n)^2 (\pi^*)^0 (\sigma^*)^0$ and that for ion OH^- is $(\sigma^n)^2 (\sigma^b)^2 (\pi^n)^4 (\sigma^*)^0$. Represent the mechanism of the first reaction step. In your analysis, specify the frontier orbitals that interact for each reactant and classify them (HOMO/LUMO/SOMO), the direction of electron transfer, the geometry of the transition state, as well as the bonds that are broken and/or formed, and the geometry of the product adduct.

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Question 3. The lattice energy of KBF_4 (10% of the total; 10th, 11th and 12th grades)

3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	Total	Total
4	4	2	8	2	2	2	10	6	40	10%

The lattice energy or net energy (U°) is the energy involved in the formation or rupture of an ionic solid, taking as reference the isolated ions (in the gaseous state). The sign of U° depends on the convention used. From now on, consider that this quantity is negative and refers to the solid formation reaction. The lattice U° values can be estimated from Coulomb's law, assuming that the interaction between the ions that form the crystal is essentially electrostatic. The Born-Mayer equation contains specific empirical-theoretical parameters that depend on both the crystalline structure (spatial arrangement of the particles) of the solid and the electronics of the ions:

$$U^\circ = -\frac{N_A \cdot M \cdot |z_+| \cdot |z_-| \cdot e^2}{4 \cdot \pi \cdot \epsilon_0 \cdot r_0} \left(1 - \frac{\rho}{r_0}\right)$$

The constant ρ has a value of 35 pm for all alkali metal halides; $N_A = 6.0221 \cdot 10^{23}$ is Avogadro's number; M is Madelung's constant and is determined by the arrangement of the ions in the crystal (crystalline structure); z_+ , z_- are the respective charges of the cation and anion; r_0 is the internuclear distance between the two nearest oppositely charged ions; $\epsilon_0 = 8.854 \cdot 10^{-12} \text{ C/(m} \cdot \text{V)}$ is the electrical permittivity in vacuum, and e is the elementary charge of the electron, whose value is $1.602 \cdot 10^{-19} \text{ C}$.

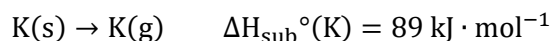
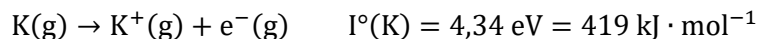
3.1. Use the previous equation to show that the U° of potassium fluoride is close to -800 kJ/mol . Consider that M is 1.7476 and the interion distance in the lattice r_0 is 271 pm. $1 \text{ pm} = 10^{-12} \text{ m}$

3.2. Usually, cations are smaller than anions, and the radii ratio is $r_+/r_- < 1$. This does not occur in the fluorides of the heavier metals. Calculate the value of r_+/r_- in potassium fluoride, knowing that the K^+ ion has an ionic radius (r_+) that is 5 pm greater than the ionic radius (r_-) of F^- .

3.3. There is a slight numerical difference between the lattice enthalpy (ΔH_U°) and U° since the latter is the variation of internal energy during the formation of the ionic solid. Although we will consider that both values are equal throughout this question, complete the following expression that relates both magnitudes for potassium fluoride by applying the First Principle of Thermodynamics.

$$\text{K}^+(\text{g}) + \text{F}^-(\text{g}) \rightarrow \text{KF}(\text{s}) \quad \Delta H_U^\circ(\text{KF}) = U^\circ(\text{KF})$$

Thermochemical cycles allow us to find energy values that are difficult to measure. For example, the Born-Haber cycle relates U° to the enthalpy of formation of the ionic solid (ΔH_f°) and different atomic and molecular properties. Consider the following data:



We will use as a convention that the electronic affinity (A° , positive) is the opposite of the enthalpy of electronic gain ($\Delta H_{\text{ge}}^\circ$, negative).

3.4. Use the Born-Haber cycle and show that A° for the fluorine atom is close to 3.5 eV.

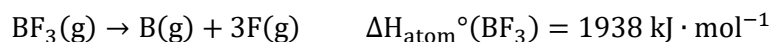
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According to Mulliken, the electronegativity (χ) equals the half-sum of the electron affinity and the ionization potential (in eV). For his part, Pearson established that hardness (η) is equal to the half-difference of the same properties to develop a mathematical basis for his Theory of Soft Hard Acids and Bases.

3.5. Calculate χ and η for the fluorine atom. $\chi(\text{F}) = \frac{I+A}{2}$ $\eta(\text{F}) = \frac{I-A}{2}$

3.6. Using your knowledge of chemical periodicity and the previous expressions, justify why fluorine is a harder species than chlorine.

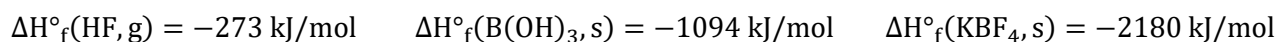
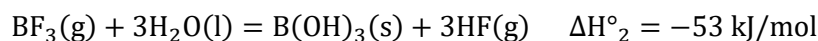
The enthalpy of atomization ($\Delta H_{\text{atom}}^\circ$) can be defined as the heat evolved (at constant temperature and pressure) in breaking all the bonds of a molecule in the gaseous state to obtain the individual atoms that make up the molecule.



3.7. Assuming that the B-F bond is purely covalent, estimate the enthalpy of formation of the gaseous tetrafluoroborate ion ΔH_1° from the above data only.



3.8. Calculate ΔH_3° for the formation of salt KBF_4 using Hess's Law and the following data.



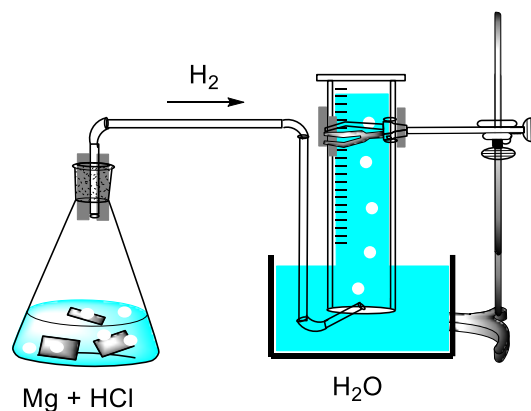
3.9. Calculate U° for KBF_4 salt employing Hess' Law and the following data. If you could not answer items 3.1, 3.7 and 3.8, you may use for your calculations the values $U^\circ(\text{KF}) = -800 \text{ kJ/mol}$, $\Delta H_1^\circ = -650 \text{ kJ/mol}$ and $\Delta H_3^\circ = -450 \text{ kJ/mol}$.



Question 4. Is it chemistry, mathematics or physics? (15% of total; 10th, 11th and 12th grades)

4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10	Total	Total
2	2	8	4	10	4	10	4	4	12	60	15%

In the chemistry laboratory of the National Training Center, a kinetic study of the reaction between magnesium and hydrochloric acid was carried out. For this purpose, the equipment shown in the figure was attached. To reduce errors, excess concentrated acid and fragments of different length l of the same tape (width $a = 3.5 \text{ mm}$; thickness $h = 0.15 \text{ mm}$) of metallic magnesium ($A_r = 24.3$; volumetric density $\rho = 1.74 \text{ g/cm}^3$ were used.). Next, the volume of dihydrogen formed as a function of time was determined for different experiments in which the variables temperature, acid concentration and exposed magnesium area were carefully controlled. The data obtained made it possible to calculate the initial reaction rates.



4.1. State the chemical equation of the reaction that occurs.

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4.2. Why do magnesium strips left outdoors turn an ashy color?

4.3. Express the rate of the reaction v_R as a function of the variation of the volume of dihydrogen formed ΔV_{H_2} in a tiny time interval Δt , the temperature T , the atmospheric pressure p_T , the vapor pressure of water p_{H_2O} and the volume of the acid solution V_D (consider this value as constant).

$$v_R = \frac{1}{2} \cdot \frac{dc(HCl)}{dt} \approx \frac{1}{2} \cdot \frac{\Delta c(HCl)}{\Delta t} \quad \text{para } \Delta t \approx 0$$

4.4. Calculate the surface densities σ (g/cm²) and linear densities λ (g/cm) of the magnesium tape used.

4.5. In some experiments, a piece of tape was used so that the total exposed magnesium surface was 70 mm². Calculate: i) the length l of the fragment, ii) the mass of magnesium it contains and iii) the volume (at 25 °C and 1.00 atm) of dihydrogen measured at the end of the reaction and the fragment being consumed.

4.6. Name all the laboratory supplies represented in the figure.

The results obtained in the experiments are summarized below:

Reaction	[H ₃ O ⁺], mol/L	Surface area of Mg, mm ²	Temperature, °C	Initial rate, mol/(L·s)
1	1,00	70	25,0	$1,56 \cdot 10^{-4}$
2	0,200	70	25,0	$5,64 \cdot 10^{-6}$
3	0,500	70	25,0	$3,36 \cdot 10^{-5}$
4	1,00	140	25,0	$2,79 \cdot 10^{-4}$
5	1,00	35	25,0	$7,30 \cdot 10^{-5}$
6	1,00	70	35,0	$2,67 \cdot 10^{-4}$
7	1,00	70	15,0	$1,19 \cdot 10^{-4}$

4.7. Determine the rate law and the constant k_R at 25°C. Round partial reaction orders to the nearest integer value. Say the units of k_R .

$$v_R = k_R \cdot A_{Mg}^{???} \cdot [H_3O^+]^{???}$$

4.8. Assuming that the activation energy is 30 kJ/mol, how many times will the rate of the reaction increase when the temperature is increased from 25 to 35 °C, keeping other variables constant?

4.9. Calculate the density (g/L) of dihydrogen at 25 °C and 1.00 atm pressure.

In their movement, the bubbles accelerate until they reach a maximum speed value when the forces are compensated. Opposing forces act on the bubbles. The thrust (E) favors the upward movement, and its module is numerically equal to the weight (F_g) of the volume of liquid displaced by the displaced bubble. Instead, the weight of the bubble and the friction force (F_d) oppose the movement. The friction force can be calculated using Stokes' Law, knowing that the dynamic viscosity coefficient of water is $\mu = 1.002 \cdot 10^{-3}$ Pa·s.

$$\text{Fuerza de gravedad } F_g = m \cdot g$$

$$\text{Ley de Stokes } F_d = 6 \cdot \pi \cdot r \cdot \mu \cdot v$$

4.10. **Find an expression** and calculate the value of the maximum velocity v_{max} (in m/s) with which dihydrogen bubbles of radius $r = 5.0$ mm move inside the container. $\rho_{H_2O} = 1.00$ g/cm³, $g = 9.81$ m/s²

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamic	$Q = \Delta U + W$
Zeroth in Celsius's scale	273,15 K	Work	$W = p \cdot \Delta V = F \cdot d$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Gibbs's energy	$\Delta G = \Delta H - T \cdot \Delta S$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{c_{\text{red}}}{c_{\text{ox}}} \right)$
Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
First order kinetics	$\ln c_t = \ln c_0 - k_r \cdot t$	Van't Hoff factor	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Cryoscopy	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbance	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Vapor pressure depression	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volume of the sphere	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Standard conditions	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Area of the sphere	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energy of a photon	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Area of the circle	$A_{\text{sp}} = \pi \cdot r^2$	Light speed	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Length of the circumference	$l_c = 2 \cdot \pi \cdot r$	Planck's constant	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Force of gravity	$F_g = m \cdot g$	Gravity acceleration on the Earth	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3		4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																		2 He 4.003
2	3 Li 6.941	4 Be 9.012												5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305												13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80	
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29	
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	*	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207,2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	**	104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]

*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]